23-27

EMBEDDED SYSTEM

PROJECT REPORT

BUTTERFLY

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Project Report

Password Based Door Locking System using 8051 Microcontroller

1. Introduction:

A Password Based Door Locking System is an electronic security system used to control access to a door using a password. Only authorized users who know the correct password can unlock the door.

In this project, an 8051 microcontroller is used as the main controller. A 4x3 matrix keypad is used to enter the password, a 16x2 LCD display shows system messages, and a DC motor simulates the door locking/unlocking mechanism. This system increases security by preventing unauthorized access.

2. Components Used:

The main components used in this project are:

- 8051 Microcontroller (AT89C51)
- 16×2 LCD Display
- 4×3 Matrix Keypad
- DC Motor (Door Lock Mechanism)
- Power Supply
- Connecting Wires
- Proteus Simulation Software
- Resistor
- LEDS

3. System Block Diagram:

The system contains the following blocks:

1. Keypad Input Unit
2. 8051 Microcontroller
3. LCD Display Unit
4. Motor Door Lock System

4. Circuit Description:

1. Microcontroller:

The 8051 microcontroller controls the entire system.It receives input from the keypad, processes the password, and controls the LCD and motor.

2. Keypad (4×3 Matrix):

The keypad is used to enter the password.

- Rows connected to Port 2
- Columns connected to Port 3

When a key is pressed, the microcontroller detects which row and column are active.

3. LCD Display:

The 16×2 LCD shows messages such as:

- ENTER PASSWORD
- ACCESS GRANTED
- WRONG PASSWORD
- TOO SHORT

LCD is connected to Port 1 for data and Port 3 for control pins (RS, RW, EN).

4. Motor (Door Lock):

The DC motor represents the door lock.

- Motor rotates when password is correct
- Motor stops when door is locked again

Motor control pins:

- P3.3 → Motor Terminal 1
- P3.4 → Motor Terminal 2

5. Software Description:

The program is written in Embedded C for 8051.

Code:

```
#include <reg51.h>
// Pins Configuration
sbit r1 = P2^0; sbit r2 = P2^1; sbit r3 = P2^2; sbit r4 = P2^3;
sbit c1 = P3^0; sbit c2 = P3^1; sbit c3 = P3^2;
sbit m1 = P3^3; sbit m2 = P3^4;
sbit rs = P3^5; sbit rw = P3^6; sbit en = P3^7;
sbit green_led = P0^0; sbit red_led = P0^1;
unsigned char master_pin[] = "12345";
unsigned char user_input[10];
unsigned char count = 0, wrong_attempts = 0;
void delay(unsigned int time) {
unsigned int i, j;
for(i=0; i<time; i++)
for(j=0; j<1275; j++);
}
```

```
void lcd_cmd(unsigned char cmd) {
P1 = cmd; rs = 0; rw = 0; en = 1; delay(2); en = 0;
}
void lcd_data(unsigned char dat) {
P1 = dat; rs = 1; rw = 0; en = 1; delay(2); en = 0;
}
void lcd_display(unsigned char *str) {
while(*str) { lcd_data(*str); str++; }
}
void lcd_init() {
lcd_cmd(0x38); lcd_cmd(0x0C); lcd_cmd(0x06); lcd_cmd(0x01); delay(5);
}
char get_key() {
char key;
while(1) {
r1=0; r2=1; r3=1; r4=1;
if(c1==0){key='1';break;} if(c2==0){key='2';break;} if(c3==0){key='3';break;}
r1=1; r2=0; r3=1; r4=1;
if(c1==0){key='4';break;} if(c2==0){key='5';break;} if(c3==0){key='6';break;}
r1=1; r2=1; r3=0; r4=1;
if(c1==0){key='7';break;} if(c2==0){key='8';break;} if(c3==0){key='9';break;}
r1=1; r2=1; r3=1; r4=0;
if(c1==0){key='*';break;} if(c2==0){key='0';break;} if(c3==0){key='#';break;}
}
```

```
while(c1==0 || c2==0 || c3==0);
delay(20);
return key;
}
void main() {
unsigned char k, i, match;
unsigned int timer;
lcd_init();
green_led = 0; red_led = 0;
while(1) {
m1 = 0; m2 = 0;
lcd_cmd(0x01);
lcd_display("ENTER PASSWORD");
lcd_cmd(0xC0);
count = 0;
while(1) {
k = get_key();
if(k == '#') break;
if(count < 5) {
user_input[count] = k;
lcd_data('*');
count++;
}
if(count == 5) break;
```

```
}

if(count == 0) continue;
if(count < 5) {
lcd_cmd(0x01);
lcd_display("TOO SHORT!");
lcd_cmd(0xC0);
lcd_display("#:RESET or WAIT");
red_led = 1;
for(timer=0; timer<3000; timer++) {
r4=0; r1=1; r2=1; r3=1; // Manual reset scanning
if(c3==0) break;
delay(1);
}
red_led = 0;
continue;
}

match = 1;
for(i=0; i<5; i++) {
if(user_input[i] != master_pin[i]) match = 0;
}

if(match == 1) {
```

```
wrong_attempts = 0;
green_led = 1;
lcd_cmd(0x01);
lcd_display("WELCOME HIRA");
lcd_cmd(0xC0);
lcd_display("DOOR UNLOCKING");
m1 = 1; m2 = 0; delay(2000);
lcd_cmd(0x01);
lcd_display("AUTO LOCKING");
m1 = 0; m2 = 1; delay(1000);
green_led = 0;
}
else {
wrong_attempts++;
red_led = 1;
if(wrong_attempts >= 3) {
lcd_cmd(0x01);
lcd_display("ACCESS DENIED!");
lcd_cmd(0xC0);
lcd_display("SYSTEM LOCKED...");
delay(10000);
wrong_attempts = 0;
}
else {
```

```
lcd_cmd(0x01);
lcd_display("WRONG PASSWORD");
lcd_cmd(0xC0);
lcd_display("#:RESET or WAIT");
for(timer=0; timer<3000; timer++) {
r4=0; r1=1; r2=1; r3=1;
if(c3==0) break;
delay(1);
}
}
red_led = 0;
}
} // End of while(1)
} // End of main
```

Main functions used in the code:

Function	Purpose
delay()	Creates delay for hardware timing
lcd_cmd()	Sends command to LCD
lcd_data()	Sends character to LCD
lcd_display()	Displays string on LCD
get_key()	Reads key pressed on keypad

The system password stored in program: **12345**

6. Working of the System:

Step 1 : System Start

When power is supplied:

- LCD displays ENTER PASSWORD
- Cursor waits for user input.

Step 2 : Password Entry

User enters password using the keypad.

Each pressed key:

- Is detected by scanning keypad rows and columns.
- Stored in the user_input array.

Entered digits are displayed as **'*'** for security.

Step 3 : Password Verification

When user presses #:

The microcontroller compares:

Entered Password
VS
Stored Password (12345)

Step 4 : Correct Password

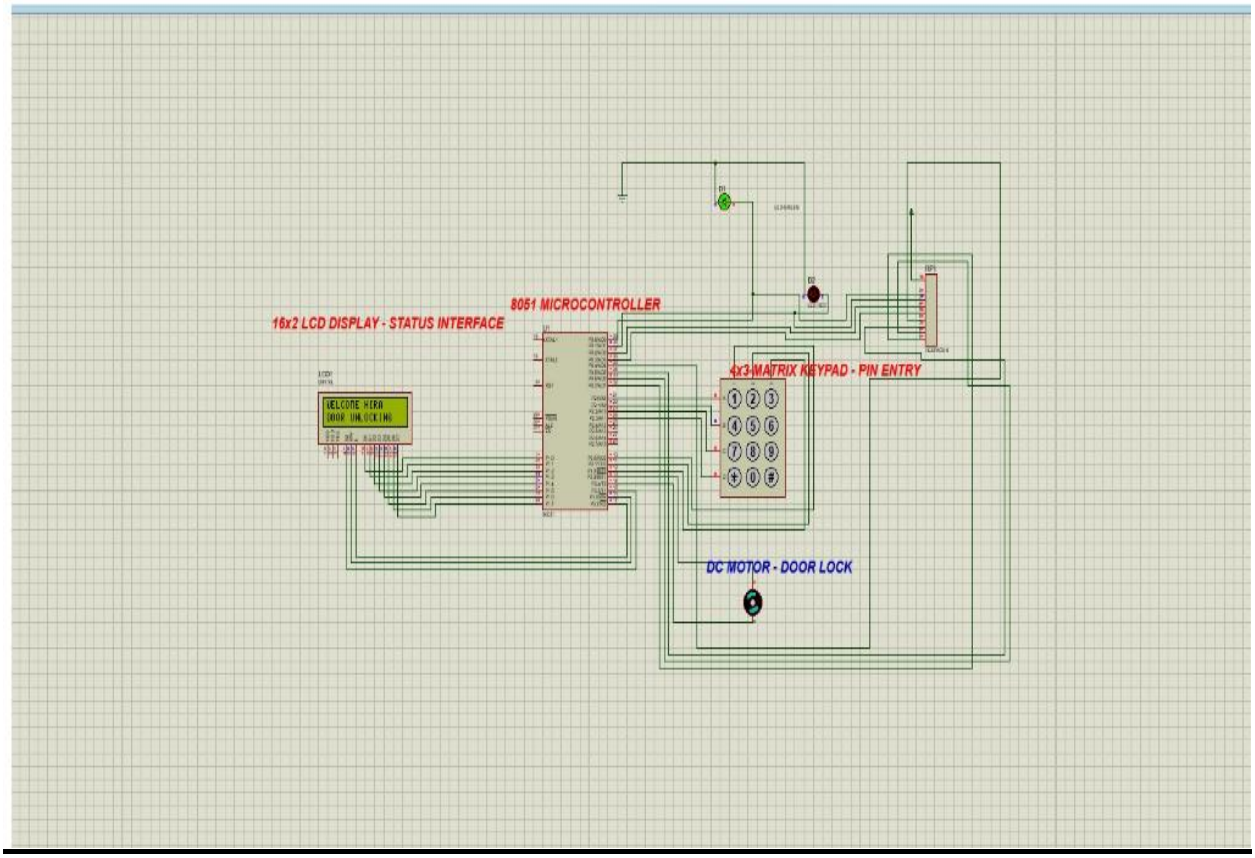
If password is correct:

LCD shows:

ACCESS GRANTED
PRESS # TO LOCK

Motor rotates to unlock the door.

Output:



Step 5 : Lock Again

When user presses # again:

- Motor stops
- System resets
- Door is locked again

Step 6 : Wrong Password

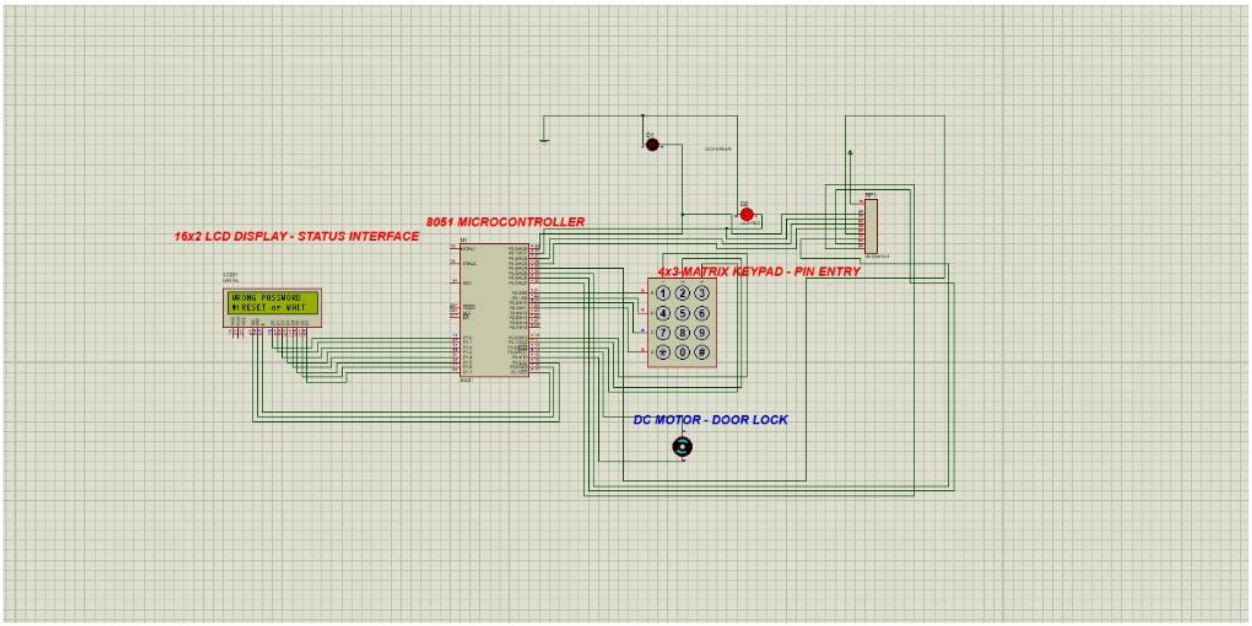
If password is incorrect:

LCD shows:

WRONG PASSWORD
PRESS # TO RESET

User must press # to try again.

Output:



Step 7 : Short Password

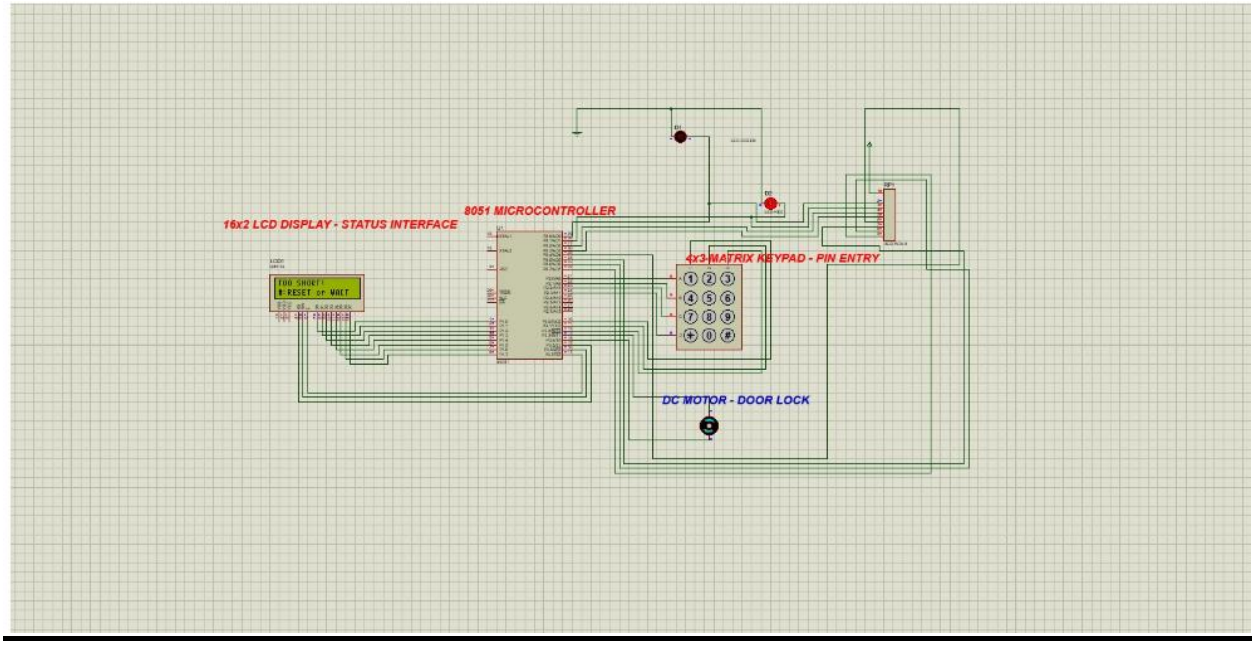
If user enters fewer than 5 digits:

LCD shows:

TOO SHORT
PRESS # TO RESET

System resets after pressing #.

Output:



7. Advantages:

- Improves security
- Easy to use
- Low cost
- Can be used in homes, offices, and lockers

8. Applications:

- Home security systems
- Office door access
- Digital lockers
- ATM security systems
- Smart door locking systems

9. Conclusion:

The Password Based Door Locking System using 8051 microcontroller provides a reliable and secure method for controlling door access. The system verifies the password entered through the keypad and allows access only to authorized users. This project demonstrates the practical use of microcontrollers in embedded security systems.